WASP-1b

R-1231

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-1_191121` · created 2026-07-31T09:56:13+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458808.77 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.035
Transit depth (Rp/Rs)^2	2e-06 +/- 0.00099 [%]
Orbital Inclination (inc)	89.74 +/- 1.43
Airmass coefficient 1 (a1)	0.85 +/- 0.41
Airmass coefficient 2 (a2)	0.576 +/- 0.089
Scatter in the residuals of the lightcurve fit is	52.86 %
Best Comparison Star	#3 - [409, 58]
Optimal Aperture	5.94
Optimal Annulus	15.6
Transit Duration (day)	0.142 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.035 vs 0.1036 (deviation 0.8σ)
Duration fit vs published	0.142 d vs 0.1567 d (deviation 1.13σ)
Mid-time O–C	-1.7 min
Depth SNR	0.0
Residual scatter	52.86 %

Light curve

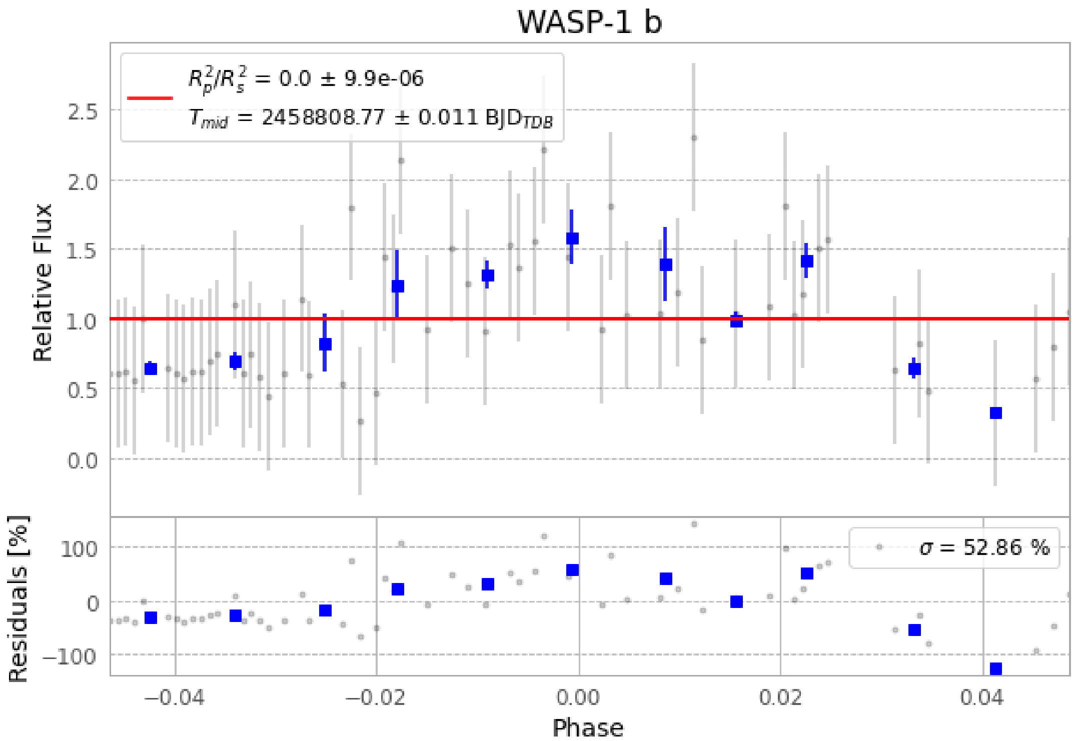

Figure 1 — FinalLightCurve_WASP-1 b_2019-11-20.png (SHA-256 2bd60c11fea3c0d6…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [258, 310], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 55ceac02c2313dca…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

117 FITS frames (77 MB), `WASP-1191121033312.FITS` → `WASP-1191121092112.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-191121.log (c9c977162414…), FinalLightCurve_WASP-1 b_2019-11-20.png (2bd60c11fea3…), FinalLightCurve_WASP-1 b_2019-11-20.csv (8c0046389e6c…)

Compute resources

Wall time 335.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)